Medication Safety Presentation for food allergies

BY OGBONNAYA OKEORJI.

Clinical Problem and Solution

Viewing a Patient's health history in addition to medication prescription to their allergy

Why it Matters

A patient's demographics such as where the patient resides may raise up questions such as: Patient A is allergic to pollen, do they live in an area susceptible to pollen (Plants, Trees, weeds, etc)

Solution:

A rule based system using predefine rules that are based on what the allergies are to accurately prescribed medication and possible other treatments

Rule Based System Demo

```python
def check_medication_safety(
    patient_allergies,
    prescribed_med,
    prescribed_treatment=None
):
    """
    Rule-Based Medication Safety Checker using if/else logic
    This is NOT AI - it follows predefined rules
    """
    alerts = []
    # Rule 1: Check peanut allergy
    if "peanut" in patient_allergies and "peanut" in prescribed_med.
    lower():
        alerts.append("Peanut Allergy detected")
        alerts.append("Recommendation: Use Epinephrine instead")

    # Rule 2: Check egg allergy
    if "egg" in patient_allergies and "egg" in prescribed_med.lower():
        alerts.append("Egg Allergy detected")
        alerts.append("Recommendation: Use loratadine instead")

    # Rule 3: Check for food allergy medications
    if "food allergy" in patient_allergies and "food allergy" in
    prescribed_med.lower():
        alerts.append("Food allergy detected")
        alerts.append("Recommendation: Use antihistamines instead")

    return alerts
```

```
Patient: Ben Tennyson
Prescribed: Epinephrine

Safety Check Results:
Treatment, emergency.
Patient: Wayne Brown
Prescribed: loratadine

Safety Check Results:
No safety alerts detected.
```

Rule 1: If patient is allergic to peanuts, treatment for epinephrine is needed, the results of this is emergency.

Rule 2: If allergic to eggs, treatment is oral medication, no safety alerts detected

How AI Would Make It Smarter

The limitation of the rule-based approach: The lack of variability!

The limit is the information already provided such as the patient history.

How can AI/machine learning improve?: providing pre assessments and a possible pre diagnosis due to other patients history having similar allergies and treatments.





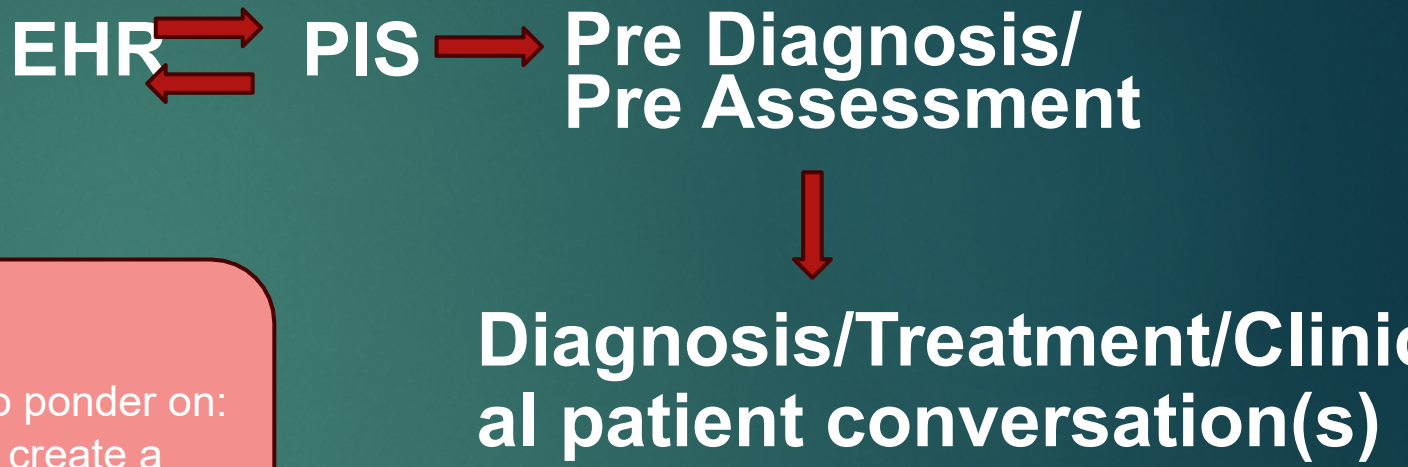
Real Life example where AI is implemented: Atopical Dermititis

Work System Analysis

(PEOPLE) Health care professionals:
Physicians
Nurses
Physician Assistants
Pharmacists,

(TASKS): Physicians and Nurses being the clinical providers to help assess and diagnose patients. Help developing and providing possible treatment plans and options to help improve patient health.

(TECHNOLOGY): A question to ponder on: How can medical providers create a healthcare web system?



Implementation Approach

- Deploy rule-based system: To identify the allergies the patient has and pre assess what medication/treatment will be used to treat/manage the allergy
- Collect Data for AI training: Collecting patient information to improve accuracy for the machine learning
- Develop AI-enhanced version: Being able to provide more information regarding treatments (we can go further: if someone was allergic to pollen, do they have air purifiers at home that can help with breathing? Preventative Health care)
- Success metrics: Improved patient care experience, more accurate diagnosis and treatments, accurate patient health records